



VEKTOR RESEARCH SERIES · 2026

Risk & Limitation Disclosure

*A structured, honest analysis of the risks and limitations of the Vektor platform —
model risk, execution risk, data risk, and operational risk —
with the mitigations in place and those still on the roadmap.*

For institutional due diligence teams, risk committees, and compliance officers

IP-WP-RISK-260501-1.0

May 2026

ABSTRACT

This paper does not exist to satisfy a legal requirement. It exists because institutional investors conducting due diligence should have a single, structured document that addresses platform risk directly — not distributed across white papers or buried in product documentation. The tone is the same as the rest of the Vektor research series: specific, honest, and free of hedging language that obscures rather than informs. Where a risk is real, this paper says so. Where a mitigation is partial, this paper says that too.

KEY TAKEAWAYS

- Four risk categories are addressed: model risk, execution risk, data risk, and operational risk. Each is assessed with mitigations in place and those still on the roadmap.
- The most significant risk at the current stage is model risk — specifically, the gap between backtest performance and live performance. This risk is acknowledged directly, with the structural mitigations the platform provides.
- Human approval gates at every capital-at-risk stage are the primary operational risk mitigation. These gates are structural — they cannot be bypassed in configuration.
- Several mitigations are on the roadmap rather than live today. This paper states which are complete and which are pending, with defined trigger conditions.
- Founding Mandate partners are invited to raise any risk not addressed here. The Founding Mandate is a partnership in which both parties are building toward the same outcome — a platform that performs as described, with the governance and controls that institutional management requires.

1. HOW TO READ THIS DOCUMENT

What this document is, what it is not, and how it relates to the rest of the Vektor research series.

This document is a practitioner's risk analysis — structured in the same format as any internal risk assessment a portfolio manager or compliance officer would produce when evaluating a new system. It identifies risks, assesses likelihood and impact, states mitigations in place, and is honest about where mitigations are still on the roadmap.

It is not a legal disclaimer. It does not attempt to limit liability through exhaustive caveats. Readers seeking a legal risk disclosure should request the separate legal documentation available to Founding Mandate partners.

The risks described in this paper are the risks of adopting a systematic portfolio management platform at an early stage of its development. They are not unique to Vektor — every systematic platform carries model risk, execution risk, data risk, and operational risk. What is specific to Vektor is how each risk is structured, mitigated, and disclosed.

2. RISK SUMMARY

Fourteen identified risks across four categories. Likelihood and impact reflect the current stage of platform development.

RISK AREA	LIKELIHOOD	IMPACT	MITIGATION IN PLACE
Backtest vs live gap	High	High	XGBoost validation on live signals. Walk-forward validation on roadmap. PM approval gate before any position is taken.
Overfitting (signal parameters)	Medium	Medium	Three-year look-back. Six-indicator grid search limits parameter space. Sharpe ratio as objective function. Disclosed in WP-04.
Signal decay	Medium	Medium	Indicators re-optimised per strategy build. Rebalancing trigger detection on roadmap. PM reviews signal performance at each approval.
Model (XGBoost) drift	Medium	Medium	Manual review when validation results appear anomalous. Automated drift detection on near-term roadmap.
Concentration risk	Medium	High	Correlation matrix verification at optimisation stage. Sector screening in universe reduction.

RISK AREA	LIKELIHOOD	IMPACT	MITIGATION IN PLACE
IBKR connectivity failure	Low	Medium	SQS queue decouples order generation from execution — orders queue, not fail. Reconnection resumes processing automatically.
Simulation vs live gap	Medium	High	Current IBKR connection is simulation account. Full order flow tested in simulation. Live execution before first Founding Mandate onboarding.
Lot rounding / mandate size	Low	Low–Med	Cash buffer absorbs rounding residual. Recommended minimum mandate size SGD 100,000+ for a ten-instrument portfolio to maintain efficiency above 99%.
Market liquidity	Low	High	Minimum average daily volume threshold applied in universe screening.
Market data outage	Low	Medium	Three years of daily history maintained in RDS. Slack alert on download failure for manual intervention.
Data quality (Yahoo Finance)	Low	Medium	Quality checks on ingestion. Corporate action adjustments applied. Institutional data feed substitution available on request.
FX rate accuracy	Low	Low	Daily FX rates fetched and stored. Position-level conversion documented in WP-05. FX source configurable.
Unauthorised action	Low	High	Three-role model enforced at application layer. Infrastructure-level RBAC on next prototype roadmap. Full audit trail of every action.
Platform unavailability	Low	High	ECS on Fargate auto-restarts failed tasks. RDS automated backups. Production observability alerting on near-term roadmap.

3. MODEL RISK

The risk that model-generated signals do not perform in live conditions as they did in backtests.

THE BACKTEST TO LIVE GAP

The most significant risk in any systematic strategy is the gap between backtest performance and live performance. Backtests are optimised on historical data. Live markets introduce conditions historical data may not have captured: regime changes, liquidity shifts, correlation breakdowns, and the market impact of the strategy itself.

Vektor's structural response is the XGBoost signal validation layer: every signal is evaluated against current market conditions before it is presented to the portfolio manager. A signal that performed well in backtests but is assessed as low-confidence against current conditions is filtered before PM review. This does not eliminate the gap — no tool does. It reduces the probability that a historically strong signal is applied in a market context where it is unlikely to perform.

The most important mitigation for model risk is structural, not technical. No Vektor strategy reaches a client portfolio without a portfolio manager reviewing the complete strategy and taking an explicit approval action. The PM's contextual judgement is the final filter that no backtest optimisation can replace.

Overfitting

The grid search optimises technical indicator parameters against a three-year historical window. Optimised parameters may reflect historical noise rather than genuine signal. Mitigations: Sharpe ratio as objective function, six-indicator universe limiting parameter space, three-year window covering multiple market regimes. Formal walk-forward validation is on the near-term roadmap and will materially strengthen this mitigation.

Signal decay

Technical indicators lose predictive power as market conditions change. Vektor's response: parameters are re-optimised at each strategy build, not fixed at initial setup. Rebalancing trigger detection is on the roadmap. Current practice: PM reviews signal performance against current market conditions at each strategy approval.

Model drift

The XGBoost model is trained on historical signal outcomes. As market conditions change, classification accuracy may degrade. Current mitigation: manual review when results appear anomalous. Automated drift detection with CloudWatch alerting is on the near-term roadmap.

Concentration risk

Efficient frontier optimisation may produce concentrated allocations in instruments with correlated historical performance. Mitigation: correlation matrix verification is a required optimisation step, flagging high pairwise correlation before PM approval. Sector diversification screening applied in universe reduction.

4. EXECUTION RISK

The risk that approved positions are not executed as intended.

Simulation vs live gap

The IBKR Gateway is currently connected to the IBKR simulation account. Simulation mode does not fully replicate live execution: fills are assumed at the requested price without market impact, partial fills do not occur, and liquidity is not modelled. The full order flow is operational and tested in simulation. Live execution will be activated before the first Founding Mandate client is onboarded — this is a defined trigger condition, not an open-ended roadmap item.

IBKR connectivity failure

If the IBKR Gateway loses connectivity, the SQS queue buffers orders rather than failing them. Orders are processed when connectivity is restored. The queue architecture was specifically designed to handle this scenario — see the decision log in WP-07. Slack notification on order completion means a failure to complete is visible immediately.

Lot rounding and minimum mandate size

Share quantities must be whole numbers. The allocation calculation rounds down to the nearest lot size, producing a residual cash balance. At 99.94%+ efficiency in simulation testing, this residual is immaterial for typical mandate sizes. The practical minimum recommended mandate size for a ten-instrument SGX portfolio is SGD 100,000 — below this threshold, lot rounding effects become more pronounced and capital allocation efficiency may fall below 98%. This threshold scales with instrument count and average lot value for other markets.

Market liquidity

Strategies that pass the liquidity screening may encounter reduced liquidity at execution time — particularly for smaller-cap instruments or in stressed market conditions. The PM review is the appropriate point to assess current liquidity conditions before approving execution.

5. DATA RISK

The risk that market data used for optimisation or execution is inaccurate, unavailable, or unrepresentative.

Market data source dependency

Equity price history and live prices are currently sourced from Yahoo Finance via API — a widely used, generally reliable source for listed equity data, but a free provider without an enterprise SLA. Data quality issues do occur occasionally. Quality checks run on ingestion. The data source is configurable: a Bloomberg or institutional market data feed can be substituted without architectural changes — available as a Founding Mandate configuration option.

Market data outage

If the daily price download fails, existing three years of historical data in RDS is unaffected. Existing approved strategies continue to function. New strategy builds or signal re-optimisations require current data and would be deferred until the feed is restored. Slack notification on download completion means failure is visible immediately.

Corporate action handling

Yahoo Finance applies adjusted close prices that account for most corporate actions. Edge cases — recent corporate actions not yet reflected in adjusted prices — may affect signal quality for affected instruments. The PM review is the appropriate point to assess any known corporate action effects on the current strategy.

FX rate accuracy

Daily FX rates are fetched and stored for all configured currency pairs. Position-level conversion uses the most recent stored rate. For portfolios with significant cross-currency exposure, intraday FX movements may introduce a small conversion variance worth monitoring at execution time.

6. OPERATIONAL RISK

The risk that platform operations introduce errors or failures that affect client outcomes.

Human error at approval gates

Every capital-affecting action requires explicit human approval — which is the primary operational risk mitigation and also the source of a residual risk: human error at the gate itself. Mitigations: the complete strategy breakdown is visible at the approval screen (instruments, weights, quantities, prices, efficiency), reducing undetected errors. The audit trail records every approval with PM identity and timestamp, supporting error investigation and correction.

Role separation

The three-role model separates research (Analyst), investment decisions (PM), and cash operations (Operations). An Analyst cannot approve strategies. Operations cannot initiate allocations. Infrastructure-level RBAC — enforcing this at the AWS IAM layer — is planned for the next prototype iteration.

Platform unavailability

ECS microservices on Fargate restart automatically on failure. RDS has automated daily backups. The SQS queue persists market orders through application restarts. No formal SLA is committed at the current stage. Production observability alerting is on the near-term roadmap. Availability commitments for Founding Mandate clients will be agreed as part of the mandate terms.

Regulatory compliance

Vektor is a portfolio construction and management platform. Regulatory obligations — licensing, client suitability, reporting — remain with the portfolio manager and the firm operating the platform. Vektor provides the audit trail and compliance documentation described in WP-06. Full details of regulatory standing are available to Founding Mandate partners on request.

7. RISK MITIGATION ROADMAP

Mitigations on the roadmap but not yet live — with defined trigger or condition for each.

MITIGATION	RISK CATEGORY ADDRESSED	TRIGGER / CONDITION
Live IBKR execution	Execution — simulation vs live gap	Before first Founding Mandate client onboards
Walk-forward validation	Model — overfitting and backtest-to-live gap	After live track record of sufficient length established
Automated drift detection	Model — XGBoost model drift	As part of production observability implementation
CloudWatch alerting	Operational — platform unavailability, data outage	Before production environment promotion
Infrastructure-level RBAC	Operational — role separation	Next prototype iteration
Bloomberg / institutional data feed	Data — market data source reliability	Available as configuration for Founding Mandate partners

8. CONCLUSION

Risk disclosure is not the same as risk management. This document is both.

The risks described in this paper are real. Some are inherent to systematic portfolio management regardless of platform. Some are specific to the current stage of Vektor's development. The mitigations described are genuine — either structurally enforced by the platform architecture or explicitly on the roadmap with defined trigger conditions.

The Founding Mandate is a partnership in which both parties are building toward the same outcome — a platform that performs as described, with the governance and controls that institutional management requires. Founding Mandate partners are invited to raise any risk they do not see addressed in this document. That conversation is part of the partnership.

For compliance and audit trail architecture, see WP-06. For AI and ML governance, see WP-08. For technical infrastructure, see WP-07. All documents at investpuppy.com.